

11.5 Mitigate STP Attacks

Mitigate STP Attacks

PortFast and BPDU Guard

Recall that network attackers can manipulate the Spanning Tree Protocol (STP) to conduct an attack by spoofing the root bridge and changing the topology of a network. To mitigate STP attacks, use PortFast and Bridge Protocol Data Unit (BPDU) Guard:

PortFast

- PortFast immediately brings a port to the forwarding state from a blocking state, bypassing the listening and learning states.
- Apply to all end-user access ports.

BPDU Guard

- BPDU guard immediately error disables a port that receives a BPDU.
- Like PortFast, BPDU guard should only be configured on interfaces attached to end devices.

Mitigate STP Attacks

Configure PortFast

PortFast bypasses the STP listening and learning states to minimize the time that access ports must wait for STP to converge.

- Only enable PortFast on access ports.
- PortFast on inter switch links can create a spanning-tree loop.

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
S1(config-if)# exit
S1(config)# spanning-tree portfast default
%Warning: this command enables portfast by default on all interfaces. You
should now disable portfast explicitly on switched ports leading to hubs,
switches and bridges as they may create temporary bridging loops.
S1(config)# exit
```

Configure PortFast (Cont.)

To verify whether PortFast is enabled globally you can use either the:

- **show running-config | begin span** command
- **show spanning-tree summary** command

To verify if PortFast is enabled an interface, use the **show running-config interface *type/number*** command.

The show **spanning-tree interface *type/number* detail** command can also be used for verification.

Mitigate STP Attacks

Configure BPDU Guard

An access port could receive an unexpected BPDUs accidentally or because a user connected an unauthorized switch to the access port.

- If a BPDU is received on a BPDU Guard enabled access port, the port is put into error-disabled state.
- This means the port is shut down and must be manually re-enabled or automatically recovered through the **errdisable recovery cause psecure_violation** global command.

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Configure BPDU Guard

BPDU Guard can be enabled:

- **On an interface** –
- Use the **spanning-tree bpduguard enable** interface configuration command.
- **Globally** –
- Use the **spanning-tree portfast bpduguard default** global configuration command to enable BPDU Guard on all access ports.

```
S1(config)# interface fa0/1
S1(config-if)# spanning-tree bpduguard enable
S1(config-if)# exit
S1(config)# spanning-tree portfast bpduguard default
S1(config)# end
S1# show spanning-tree summary
Switch is in pvst mode
Root bridge for: none
Extended system ID          is enabled
Portfast Default             is enabled
PortFast BPDU Guard Default is enabled
Portfast BPDU Filter Default is disabled
Loopguard Default           is disabled
EtherChannel misconfig guard is enabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short
(output omitted)
S1#
```